

VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN
(Course Code: ECE 1003)

Module-1:Lecture-11

Boolean Algebra

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Module-1 (Part-2)

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BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Gate Level Minimization

GATE LEVEL MINIMIZATION

- ✓ **Gate-level minimization** is the design task of finding **an optimal gate-level implementation** of the Boolean functions describing a digital circuit.
- ✓ As discussed earlier that algebraic method is not sufficient enough to find out whether a given expression can be simplified or not.
- ✓ To avoid the above problem, **Karnaugh's Map or K-map method** is developed to simplify a Boolean expression.
- ✓ It is a **graphical representation** of Boolean algebra.
- ✓ The simplified expressions produced by the map are always in one of the two standard forms: **sum of products or product of sums**.
- ✓ This is another way to represent the truth table in which **each block** represents **one minterm**.

2-VARIABLE K-MAP

m_0	m_1
m_2	m_3

		y'		y	
		0		1	
x'	0	m_0 $x'y'$	m_1 $x'y$		
	1	m_2 xy'	m_3 xy		

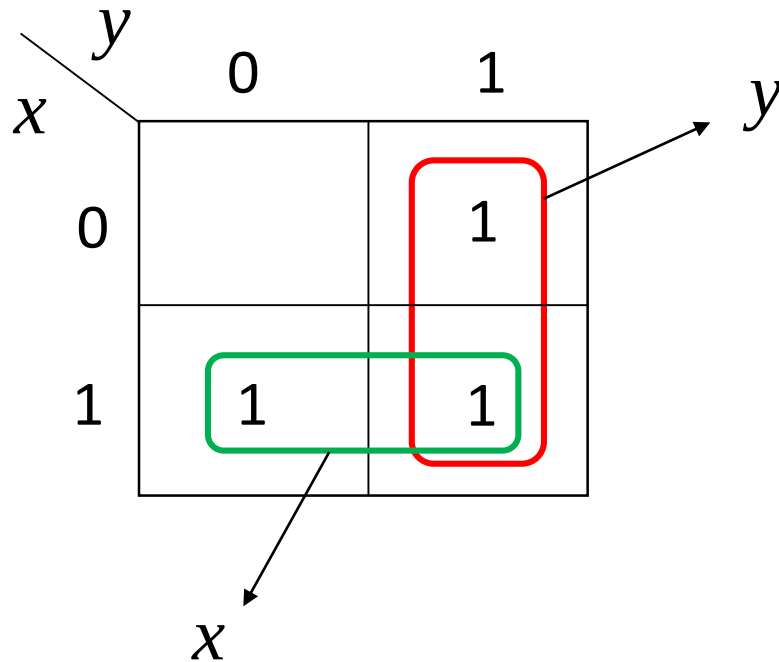
Any **two adjacent squares** in the map **differ by only one variable**, which is primed in one square and unprimed in the other.

2 Boolean variable $\longrightarrow$ Map contains 4 squares

One squares $\longrightarrow$ One minterm

2-VARIABLE K-MAP

Example: Simplify $F = \Sigma(1, 2, 3)$

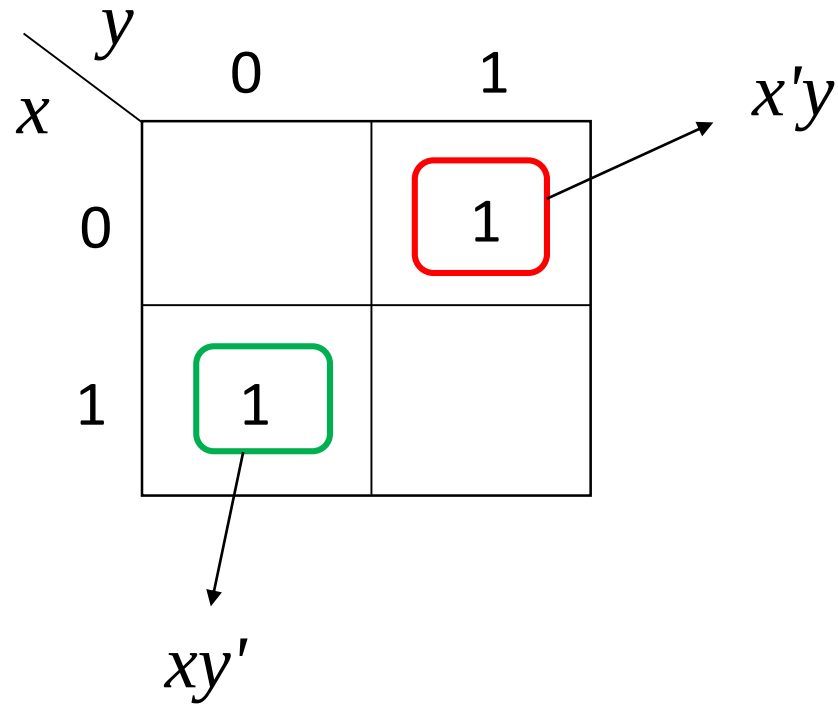


- ✓ Minterms 1 and 3 are **paired** in which only x is differed, thus, removed and the term becomes y .
- ✓ Minterms 2 and 3 are **paired** in which only y is differed, thus, removed and the term becomes x .

$$F = x + y$$

2-VARIABLE K-MAP

Example: Simplify $F = \Sigma(1, 2)$



✓ The minterms does not make any pair or quad so can not be minimized.

$$F = x'y + xy'$$

2-VARIABLE K-MAP

Example: Simplify $F = \Sigma(0, 2)$

		y	
		0	1
x	0	1	
	1	1	

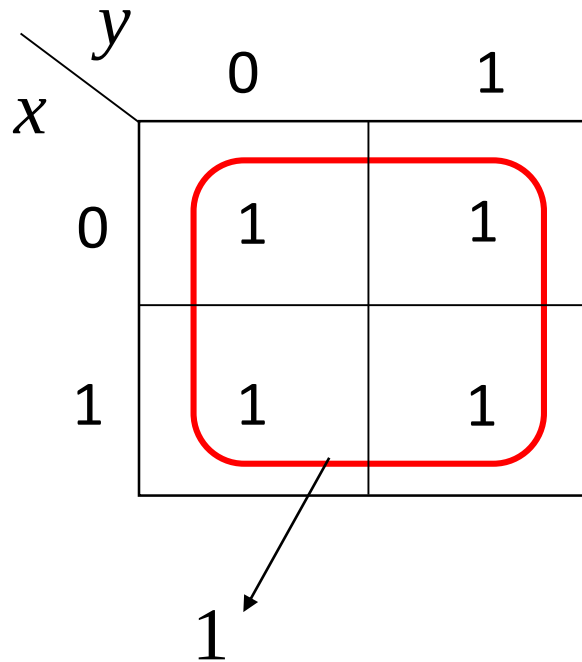
y'

- ✓ Minterms 0 and 2 are **paired** in which only x is differed, thus, removed and the term becomes y' .

$$F = y'$$

2-VARIABLE K-MAP

Example: Simplify $F = \Sigma(0, 1, 2, 3)$



- ✓ Minterms 0, 1, 2 and 3 formed **a quad** in which both x and y is differed, thus, both can be removed, and the result becomes **1**.

$$F = 1$$

3-VARIABLE K-MAP

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6

		y			
		yz			
		00	01	11	10
x	0	m_0 $x'y'z'$	m_1 $x'y'z$	m_3 $x'yz$	m_2 $x'yz'$
	1	m_4 $xy'z'$	m_5 $xy'z$	m_7 xyz	m_6 xyz'

3 Boolean variable $\longrightarrow$ Map contains 8 squares

- ✓ The minterms are arranged, **not in a binary sequence**, but in a **sequence similar to the Gray code**. The characteristic of this sequence is that **only one-bit changes in value from one adjacent column to the next**.

3-VARIABLE K-MAP

Example: Simplify $F = \Sigma(2, 3, 4, 5)$

		yz				
		00	01	11	10	
x	0			1	1	$x'y$
	1	1	1			

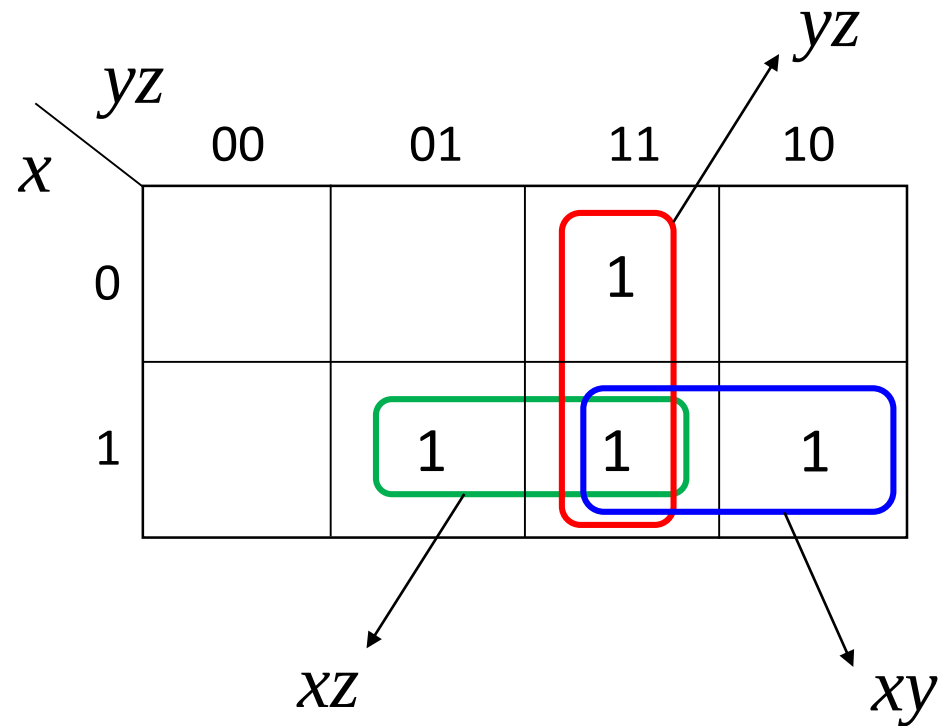
xy'

- ✓ Minterms 2 and 3 are **paired** in which only z is differed, thus, removed and the term becomes $x'y$.
- ✓ Minterms 4 and 5 are **paired** in which only z is differed, thus, removed and the term becomes xy' .

$$F = x'y + xy'$$

3-VARIABLE K-MAP

Example: Simplify $F = \Sigma(3, 5, 6, 7)$



- ✓ Minterms 3 and 7 are **paired** in which only **x** is differed, thus, removed and the term becomes **yz**.
- ✓ Minterms 5 and 7 are **paired** in which only **y** is differed, thus, removed and the term becomes **xz**.
- ✓ Minterms 6 and 7 are **paired** in which only **z** is differed, thus, removed and the term becomes **xy**.

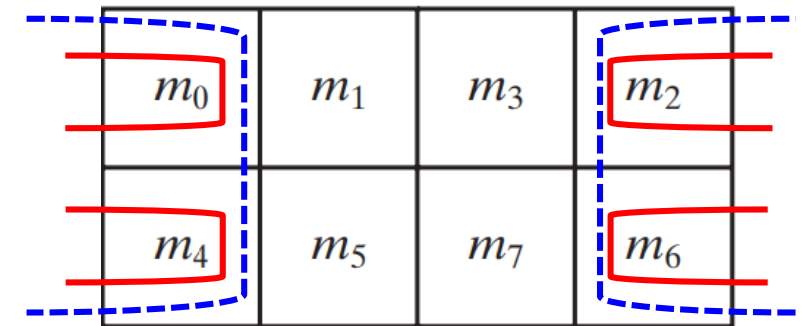
$$F = xy + yz + xz$$

3-VARIABLE K-MAP

Note:

- ✓ In certain cases, two squares in the map are considered to be adjacent even though they don't touch each other.

m_0 & m_2 } Minterms are differed by only one variable
 m_4 & m_6 } i.e., $m_0 + m_2 = x'z'$ & $m_4 + m_6 = xz'$



- ✓ Any combination of **4 adjacent squares** in the 3-variable map i.e., logical sum of 4 minterms **differed by 2 literals**.

m_0, m_1, m_4, m_5
 m_1, m_3, m_5, m_7
 m_2, m_3, m_6, m_7 } Quads

m_0, m_1, m_2, m_3
 m_4, m_5, m_6, m_7 } Bars

m_0, m_2, m_4, m_6 Quad

3-VARIABLE K-MAP

Example: Simplify $F = \Sigma(0, 1, 4, 5)$

		yz			
		00	01	11	10
x	0	1	1		
	1	1	1		

y'

- ✓ Minterms 0, 1, 4 and 5 formed **a quad** in which both x and z are differed, thus, both can be removed, and the result becomes y' .

$$F = y'$$

3-VARIABLE K-MAP

Example: Simplify $F = \Sigma(0, 1, 2, 3, 5, 7)$

yz		x'			
		00	01	11	10
x	0	1	1	1	1
	1		1	1	

Diagram illustrating the 3-variable K-map for $F = \Sigma(0, 1, 2, 3, 5, 7)$. The map is a 2x4 grid with rows labeled x (0, 1) and columns labeled yz (00, 01, 11, 10). The map shows the following values:

- Row $x=0$: $yz=00$ is 1, $yz=01$ is 1, $yz=11$ is 1, $yz=10$ is 1.
- Row $x=1$: $yz=00$ is 0, $yz=01$ is 1, $yz=11$ is 1, $yz=10$ is 0.

Groupings are shown:

- A green rectangle (quad) groups the four 1s in the top row ($x=0$), which simplifies to z .
- A red rectangle (bar) groups the four 1s in the middle columns ($yz=01, 11$), which simplifies to x' .

- ✓ Minterms 1, 3, 5 and 7 formed **a quad** in which both x and y are differed, thus, both can be removed, and the result becomes z .
- ✓ Minterms 0, 1, 2 and 3 formed **a bar** in which both y and z are differed, thus, removed and the term becomes x' .

$$F = x' + z$$

3-VARIABLE K-MAP

Example: Simplify the Boolean function using K map,

$$F = A'C + A'B + AB'C + BC$$

Solution:

$$F = A'C + A'B + AB'C + BC$$

$$= A'(B + B')C + A'B(C + C') + AB'C + (A + A')BC$$

$$= A'BC + A'B'C + A'BC + A'BC' + AB'C + ABC + A'BC$$

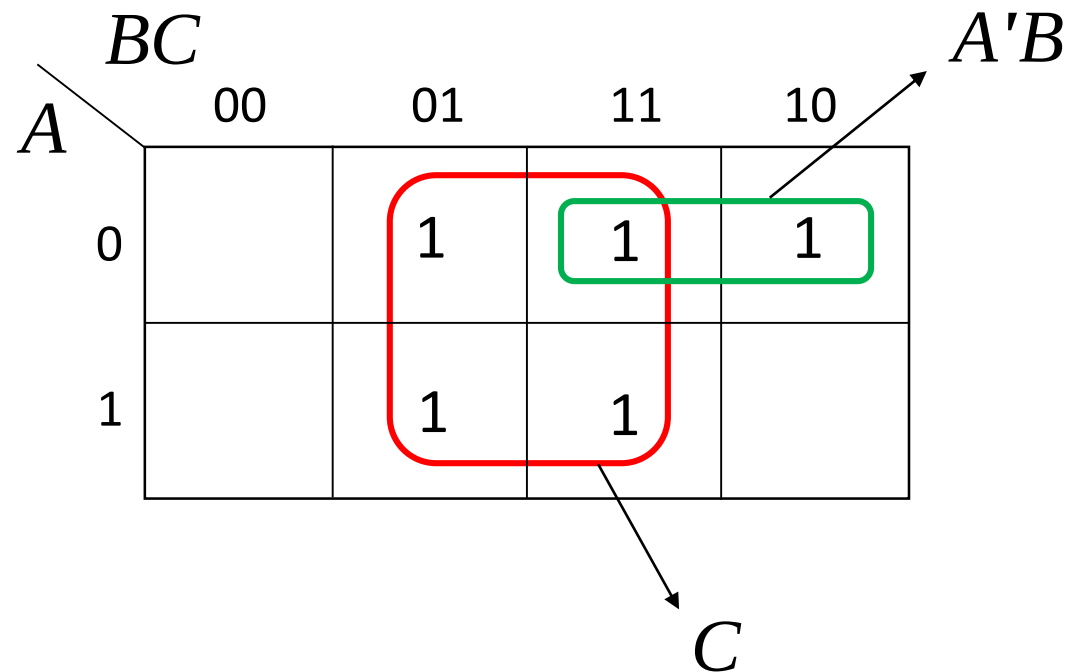
$$= A'BC + A'B'C + A'BC + A'BC' + AB'C + ABC + A'BC$$

$$= A'BC + A'B'C + A'BC' + AB'C + ABC$$

$$= m_3 + m_1 + m_2 + m_5 + m_7 = \Sigma (1, 2, 3, 5, 7)$$

3-VARIABLE K-MAP

Example: now simplify the Boolean function using K map, $F = \Sigma (1, 2, 3, 5, 7)$

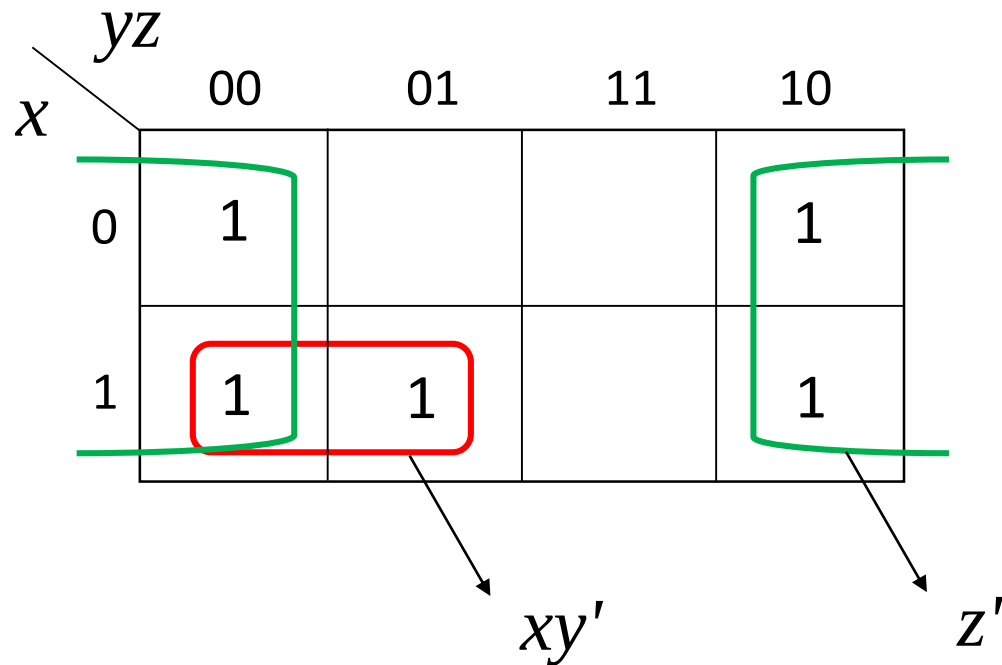


- ✓ Minterms 1, 3, 5 and 7 formed a **quad** in which both **A** and **B** are differed, thus, both can be removed, and the result becomes **C**.
- ✓ Minterms 2 and 3 are **paired** in which only **C** is differed, thus, removed and the term becomes **A'B**.

$$F = C + A'B$$

3-VARIABLE K-MAP

Example: Simplify $F = \Sigma(0, 2, 4, 5, 6)$

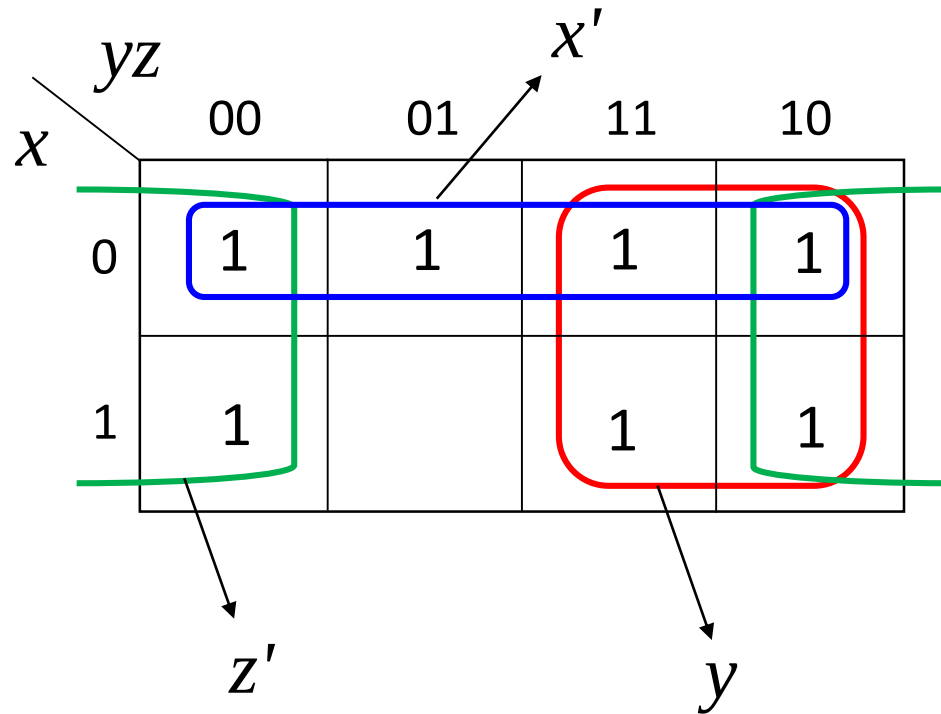


- ✓ Minterms 0, 2, 4 and 6 formed a **quad** in which both x and y are differed, thus, both can be removed, and the result becomes z' .
- ✓ Minterms 4 and 5 are **paired** in which only z is differed, thus, removed and the term becomes xy' .

$$F = z' + xy'$$

3-VARIABLE K-MAP

Example: Simplify $F = \Sigma(0, 1, 2, 3, 4, 6, 7)$



- ✓ Minterms 0, 2, 4 and 6 formed **a quad** in which both **x** and **y** are differed, thus, both can be removed, and the result becomes **z'**.
- ✓ Minterms 0, 1, 2 and 3 formed **a bar** in which both **y** and **z** are differed, thus, removed and the term becomes **x'**.
- ✓ Minterms 2, 3, 6 and 7 formed **a quad** in which both **x** and **z** are differed, thus, both can be removed, and the result becomes **y**.

$$F = x' + y + z'$$

3-VARIABLE K-MAP

Example: Simplify $F = \Sigma(0, 1, 2, 3, 4, 5, 6, 7)$

		yz			
		00	01	11	10
x	0	1	1	1	1
	1	1	1	1	1

1

✓ Minterms 0, 1, 2, 3, 4, 5, 6, 7 form **an octet**, thus the result becomes 1.

$$F = 1$$

Thank You